

MS Bioengineering

Awarded by: SCTIMST. Trivandrum.

The Department of Bioengineering at Christian Medical College (CMC), Vellore, is an academic engineering department that works closely with several clinical and basic science departments at CMC to develop engineering solutions to healthcare problems. The department's current research and development activities span medical robotics, movement analysis, physiological measurements, biomedical signal processing, and machine learning. It offers several degree programmes and lecture- and laboratory-based courses in Transducers and Instrumentation, Imaging Systems, Physiological Systems, Signal Processing and Data Analysis, Medical Robotics, and related areas. The strong, clinically grounded, interdisciplinary research at the department has resulted in several affordable, relevant solutions for the Indian healthcare system. Examples include the Bombay-Vellore artificial limb (with the Dept. of Physical Medicine and Rehabilitation (PMR)), the CMCDQA data acquisition system (with the Dept. of Physiology), the PLTUO portable hand therapy robot (with the Depts. of Neurology and PMR), and the UroD portable urodynamics machine (with the Dept. of PMR).

The MS Bioengineering is a specialized, research-oriented engineering programme conducted at the Department of Bioengineering, CMC Vellore, under the Sree Chitra Tirunal Institute of Medical Sciences and Technology (SCTIMST), Thiruvananthapuram, which awards the degree. SCTIMST is an Institute of National Importance under the Government of India. Candidates in this programme undertake a year-long research project, working closely with at least one clinical or basic science department to formulate a research question. These projects often progress to clinical trials involving human subjects, highlighting the strong clinical component of the programme and making it a distinctive feature of the MS Bioengineering degree. The programme equips students to apply quantitative and engineering methods to solve healthcare problems, preparing them for careers in research and development in the medical devices industry, as well as in academic research and teaching.

Eligibility:

All aspiring candidates should have an undergraduate degree (BTech/BE) in engineering, preferably Electrical, Electronics, Mechanical, Computer Science or equivalent. Students with an undergraduate degree (BTech/BE) in Biotechnology, Bioinformatics, Biochemical Engineering, etc. will not be eligible for the course.

Admissions:

Please keep checking our website for the admission schedule. A written test and interview of short-listed candidates for the MS programme will be conducted to select suitable candidates.

M.Tech. Clinical Engineering and PhD. Medical Devices

Awarded by: IIT-Madras & SCTIMST, Thiruvananthapuram

The MTech. and PhD programmes are jointly conducted by the three institutions, Indian Institute of Technology (IIT) – Madras, Sree Chitra Tirunal Institute for Medical Sciences and Technology – Thiruvananthapuram, and Christian Medical College – Vellore, with students spending prescribed durations in each institution. The degree is awarded by IIT Madras. All candidates must have an engineering undergraduate degree. The aim of the M.Tech. Clinical Engineering programme is to train engineers to design and manage instruments and ensure the safe and effective use of technology in hospitals and healthcare delivery points.

The MTech. programme is 2 years in duration. In the first semester of the programme, the students are at IIT Madras where they take courses in engineering. In the second semester, the students are at CMC Vellore taking courses in engineering, Human physiology, and anatomy. During this period, they also do multiple clinical visits and do a clinical project. The third semester is at SCTIMST, undergoing courses in medical devices and biomedical technology. In the fourth semester, a project is undertaken by students at their chosen institute.

The PhD Biomedical Devices program has basic courses at IIT-Madras and CMC-Vellore to prepare them for research with a clinical orientation. Students will select an area of research and conduct research with supervisors in at least two out of the three institutions.

Eligibility:

Please refer to the IIT Madras MTech admissions and PhD (Research) admissions page. Admissions are administratively handled by the office of the IIT- Madras and selection is by a committee comprising members from all three institutions. Please keep checking our website or IIT Madras' for the admission schedule.